Yitian Zhang

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EDUCATION

McGill University — B.Sc., Joint Honours in Mathematics and Computer Science — Expected April 2028

- CGPA: 3.96 / 4.00
- Placed out of Algebra 3 (MATH 456) through self-study.
- Graduate coursework (2026–27): five graduate-level mathematics courses — Advanced Real Analysis (MATH 564/565), Higher Algebra (MATH 570/571), Descriptive Set Theory (MATH 592).
- Independent study: functional analysis (MIT 18.102, completed), complex analysis (Ahlfors), abstract algebra (Artin).

Marianopolis College — DEC, Pure and Applied Science — 2023 – 2025

AWARDS

Mackey-Glass Summer Research Bursary — McGill University — 2026

RESEARCH EXPERIENCE

Undergraduate Researcher — Chacron Lab, Department of Physiology, McGill University — Summer 2026

- Built a recurrent excitatory–inhibitory spiking network model (exponential integrate-and-fire neurons) of gamma-band (30–58 Hz) oscillations in vestibular cortex (PIVC) local field potentials.
- Reproduced the key experimental finding — naturalistic vestibular stimulation evokes gamma while sinusoidal stimulation does not — and identified an inhibition-driven (ING) mechanism gated by the interneuron operating point.
- Diagnosed a model–data coherence gap by systematic hypothesis elimination across 14+ model iterations; validated the analysis pipeline against the experimental collaborator's coherence code.
- Archived code and data to the lab's Borealis repository; manuscript in preparation with Dr. Maurice Chacron.

TEACHING EXPERIENCE

Peer Tutor / Alumni Tutor — Marianopolis College, Montreal, QC — Jan 2025 – Present

- Tutor students across ten courses: Calculus 1–3, Linear Algebra 1–2, Probability & Statistics, Mechanics, Electricity & Magnetism, Waves/Optics/Modern Physics, General Chemistry, and Chemistry of Solutions.
- Write original LaTeX problem sets with worked solutions and answer keys (e.g., Calculus 2 integration techniques).

Private STEM & Mandarin Tutor — Montreal, QC / remote — Sep 2023 – Present

- Tutored students from grade 6 through CEGEP in mathematics (up to differential calculus and probability), chemistry, physics, and Mandarin (HSK 1–2), in person and online.
- Planned lesson materials and rubrics in advance; monitored progress and reported feedback to parents.

PROJECTS

Constrained Shortest-Path Solver — Java — 2025

- Implemented Dijkstra's algorithm with Lagrangian relaxation (LARAC) to compute shortest paths under edge constraints for a grid-world pathfinding course project.

OTHER EXPERIENCE

Badminton Coach — Yangtze International College & private clients, Montreal, QC — Dec 2022 – Jan 2024

- Coached technique and tactics for youth players (ages 10–12) and beginners; designed progressive training plans and scheduled facilities.

SKILLS & CERTIFICATIONS

- **Programming:** Python, Java, C, OCaml (functional programming), shell scripting, Git, LaTeX
- **Languages:** Mandarin (native), English (fluent)
- **Certifications:** First Aid (2023)